Supplementary Analysis: Baseline Age as Predictor

Does the FRS 3-way interaction reflect age per se?

Sofia Fernandez-Lozano

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1 Rationale

The primary manuscript shows that FRS produces significant 3-way interactions (YRS × CVR × HVR z-score) and LGCM a-paths, whereas age-adjusted $\text{CVR}_{\text{mimic}}$ does not. The most parsimonious explanation is that FRS effects are carried by the age component of the Framingham equation. To test this directly, we replaced the CVR predictor with z-scored baseline age (Age_{bl}) and re-ran both the LME and LGCM analyses.

If FRS effects are purely age-driven, Age_{bl} should recapitulate the FRS pattern (same domains significant, similar effect sizes). If FRS captures something beyond age alone (e.g., age × risk factor interactions embedded in the Framingham algorithm), the patterns should diverge.

1.1 Model specification

Because age cannot simultaneously serve as predictor and covariate, Age_{bl} was dropped from the covariate set when it served as the 3-way interaction predictor:

- **FRS/$\mathbf{CVR_{mimic}}$ models:** $\text{Cog} \sim \text{YRS} \times \text{CVR} \times \text{HVR}_{\text{z}} + \text{I}(\text{YRS}^2) \times \text{CVR} \times \text{HVR}_{\text{z}}$
  $- \text{Age}_{\text{bl}} + \text{EDUC} + \text{APOE4} + (\text{YRS} \mid \text{PTID})$
- **$\mathbf{Age_{bl}}$ model:** $\text{Cog} \sim \text{YRS} \times \text{Age}_{\text{bl,z}} \times \text{HVR}_{\text{z}} + \text{I}(\text{YRS}^2) \times \text{Age}_{\text{bl,z}} \times \text{HVR}_{\text{z}} + \text{EDUC} + \text{APOE4} + (\text{YRS} \mid \text{PTID})$

This difference in covariate structure means the β coefficients are not directly comparable across models (they are conditioned on different covariate sets). Accordingly, no formal bootstrap $\Delta\beta$ test was performed. The comparison is qualitative: we examine whether the same domains reach significance and whether effect directions are consistent.

For the LGCM, the same logic applies: the age covariate paths ($\text{AGE}_{\text{c}} \rightarrow$ latent intercepts and slopes) were removed when AGE_{c} served as the a-path predictor, since the mediation and predictor paths already capture all age effects on the latent factors.

2 LME Results

Table 1: **LME 3-way interaction: $\mathbf{Age_{bl}}$ vs FRS vs $\mathbf{CVR_{mimic}}$.** Sex-stratified results for the YRS $\times$ Predictor $\times$ HVR z-score interaction term. FDR correction applied within each predictor (6 tests: 3 domains $\times$ 2 sexes). Note: Age_{bl} models exclude Age_{bl} from covariates; FRS and CVR_{mimic} models include Age_{bl} as covariate.

Predictor	Sex	Domain	β	SE	p	p_{FDR}	
FRS	Male	Executive Function	-0.0198	0.0049	< .001	< .001	***
FRS	Male	Language	-0.0147	0.0045	0.001	0.002	***
FRS	Male	Memory	-0.0112	0.0038	0.004	0.005	***
CVR_mimic	Male	Executive Function	-0.0067	0.0050	0.179	0.538	
CVR_mimic	Male	Language	0.0004	0.0046	0.928	0.928	
CVR_mimic	Male	Memory	0.0005	0.0039	0.906	0.928	
Age_bl	Male	Executive Function	-0.0060	0.0056	0.284	0.385	
Age_bl	Male	Language	-0.0051	0.0051	0.321	0.385	
Age_bl	Male	Memory	0.0009	0.0044	0.829	0.829	
FRS	Female	Executive Function	-0.0597	0.0162	< .001	< .001	***
FRS	Female	Language	-0.0405	0.0152	0.008	0.009	***
FRS	Female	Memory	-0.0267	0.0128	0.038	0.038	***
CVR_mimic	Female	Executive Function	-0.0226	0.0086	0.008	0.051	*
CVR_mimic	Female	Language	-0.0035	0.0079	0.657	0.928	
CVR_mimic	Female	Memory	0.0037	0.0066	0.575	0.928	
Age_bl	Female	Executive Function	-0.0251	0.0072	< .001	0.003	***
Age_bl	Female	Language	-0.0187	0.0068	0.006	0.018	***
Age_bl	Female	Memory	-0.0064	0.0057	0.258	0.385	

Males: FRS is significant in 3/3 domains, Age_{bl} in 0/3, CVR_{mimic} in 0/3. The Age_{bl} pattern does **not** mirror FRS in males — FRS is significant across all domains but Age_{bl} is not significant in any.

Females: FRS is significant in 3/3 domains, Age_{bl} in 2/3, CVR_{mimic} in 1/3. Age_{bl} mirrors FRS in Language and Executive Function but not Memory.

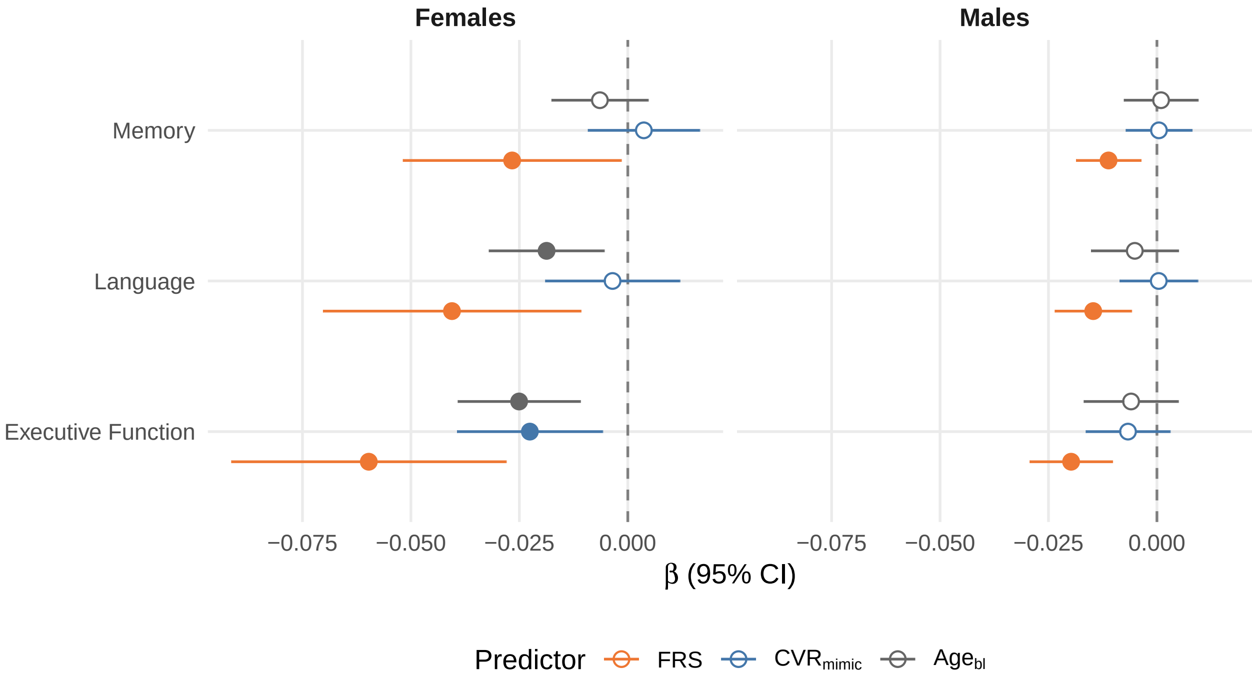

Figure 1: **LME 3-way interaction by predictor.** Comparison of FRS, CVR$_{\text{mimic}}$, and Age$_{\text{bl}}$ (z-scored baseline age) across sex and domain. Note different covariate sets: Age$_{\text{bl}}$ models exclude Age$_{\text{bl}}$ from covariates.

3 LGCM Mediation Results

Table 2: **LGCM mediation paths: Age$_{\text{bl}}$ vs FRS.** A-path (predictor → HVR slope), b-path (HVR slope → cognitive slope), indirect effect (Sobel test), and c' (direct effect). Sex-stratified, EDT time metric. Age$_{\text{bl}}$ models use Wald/Sobel inference only (no bootstrap). Female Age$_{\text{bl}}$ models converged with optimizer warnings (status RED) due to near-zero a-path estimates.

Predictor	Sex	Domain	a	p_a	b	p_b	ab	p_{Sobel}	c'	p~c'~
FRS	Female	Executive Function	0.03235	0.017	0.1640	0.007	0.005304	0.073	-0.00255	0.782
FRS	Female	Language	0.03142	0.020	0.0588	0.238	0.001847	0.293	-0.00034	0.964
FRS	Female	Memory	0.03087	0.023	0.1458	0.004	0.004501	0.075	-0.00029	0.975
Age_bl	Female	Executive Function	-0.00013	0.896	0.1578	0.008	-0.000020	0.896	0.00224	< .001
Age_bl	Female	Language	-0.00017	0.860	0.0581	0.236	-0.000010	0.861	0.00056	0.318
Age_bl	Female	Memory	-0.00016	0.867	0.1466	0.003	-0.000024	0.867	0.00033	0.605
FRS	Male	Executive Function	0.02113	0.001	0.1268	0.003	0.002679	0.028	0.00112	0.775
FRS	Male	Language	0.02132	0.001	0.0717	0.085	0.001529	0.128	0.00063	0.870
FRS	Male	Memory	0.02087	0.001	0.1430	< .001	0.002986	0.013	0.00182	0.616
Age_bl	Male	Executive Function	-0.00446	< .001	0.1302	0.002	-0.000581	0.009	0.00376	< .001
Age_bl	Male	Language	-0.00446	< .001	0.0770	0.058	-0.000343	0.079	0.00185	< .001
Age_bl	Male	Memory	-0.00448	< .001	0.1540	< .001	-0.000689	0.001	0.00199	< .001

Males: Age$_{\text{bl}}$ a-path significant in 3/3 domains (all $p < .001$), matching the FRS pattern (3/3). Age predicts hippocampal decline in males, consistent with FRS effects being age-driven.

Females: Age$_{\text{bl}}$ a-path significant in 0/3 domains, while FRS is significant in 3/3. The models converged with optimizer warnings (status RED), but this is a numerical artifact of near-zero a-path estimates producing a flat likelihood surface — all parameter estimates and standard errors are well-behaved. Age does not predict hippocampal decline in females despite FRS doing so, suggesting FRS captures something beyond chronological age in this group.

4 Summary

Table 3: **Pattern comparison across predictors.** Number of significant effects (uncorrected $p < .05$) out of possible 3 domains per sex. LME = 3-way interaction; LGCM = a-path (predictor → HVR slope).

Analysis	Predictor	Sex	Sig / 3
LME 3-way	FRS	Male	3 / 3
LME 3-way	FRS	Female	3 / 3
LME 3-way	CVR_mimic	Male	0 / 3
LME 3-way	CVR_mimic	Female	1 / 3
LME 3-way	Age_bl	Male	0 / 3
LME 3-way	Age_bl	Female	2 / 3
LGCM a-path	FRS	Male	3 / 3
LGCM a-path	FRS	Female	3 / 3
LGCM a-path	CVR_mimic	Male	0 / 3
LGCM a-path	CVR_mimic	Female	0 / 3
LGCM a-path	Age_bl	Male	3 / 3
LGCM a-path	Age_bl	Female	0 / 3

5 Interpretation

The Age_{bl} analysis yields a **partial, sex-dependent recapitulation** of the FRS pattern:

1. **Male LGCM a-path:** Age_{bl} mirrors FRS (3/3 significant). Age predicts hippocampal decline, consistent with FRS effects being age-driven in males.
2. **Male LME 3-way:** Age_{bl} does **not** mirror FRS (0/3 vs 3/3). The 3-way moderation (how age × HVR coupling changes over time) differs from the FRS pattern, despite the a-path match.
3. **Female LME 3-way:** Age_{bl} partially mirrors FRS (2/3 vs 3/3), matching in Language and Executive Function but not Memory.
4. **Female LGCM a-path:** Age_{bl} does **not** mirror FRS (0/3 vs 3/3). Age does not predict hippocampal decline in females, suggesting FRS captures age × risk factor interactions beyond chronological age alone.

5.1 Caveats

- **Different covariate structures** preclude formal β comparison. When Age_{bl} is the predictor, it is dropped as a covariate. FRS/$\text{CVR}_{\text{mimic}}$ models retain Age_{bl} as a covariate. This means the coefficients are conditioned on different covariate sets.
- **The incomplete overlap** is itself informative: FRS captures more than age per se. The Framingham equation embeds age × risk factor interactions (e.g., age modifies the weight given to smoking, diabetes, blood pressure) that are not reducible to chronological age alone.
- **Female LGCM convergence warnings** are benign numerical artifacts of near-zero a-path estimates, not model misspecification.

5.2 Recommendation

The Age_{bl} results strengthen the manuscript's central argument (FRS effects are age-confounded) but with important nuance: FRS is not *purely* age. Consider including as a supplementary table/figure if reviewers request evidence for the age-confounding claim, with appropriate caveats about covariate structure differences.